LEF MODEL · CHAPTER 15

Chapter 15: The South Pole–Aitken Basin (SPA) — the Subsurface Burner, Two Foci and Two Crustal Rings

Based on Paper 08 — South Pole–Aitken: The Crater-Poor Centre (Parts A–B9: the 5 studies and the anomalies B1–B9)

01 · THE MYSTERY

The mystery: why does the oldest and deepest basin on the Moon have a centre empty of craters — and high rims?

On the far side of the Moon, hidden from our eyes, lies **the South Pole–Aitken basin (SPA)** — the largest, deepest and oldest known basin on the Moon's face: a huge ellipse measuring 2,400 × 2,050 km (about the distance from Paris to Moscow), whose floor lies 6–9 km below the surrounding highlands and whose age is dated to 4.25 billion years. By all the rules of planetary geology, such an ancient surface, exposed to the sky for more than four billion years, must be crowded with craters. But careful measurement reveals a contradictory riddle:

1. **A centre empty of craters that grows sharper the more the ground is cleaned (B1):** when the lava plains, the 34 maria and a 200-km halo around each mare are removed from both sides, it turns out that in the central ring (0°–10°) only **125.7 craters** (20+ km) per million km² survived — only **0.51** of the density of the ring outside the rim (247.5), and at the minimum point (45.4°S, 197.9°E, about 270 km north-north-east of the centre, **exactly on**

the longitude of the antipode of Mare Serenitatis!) the density drops to 111 (0.45).

2. **A selective disappearance of the intermediate craters, and shallow floors under high rims:** the small craters (10–20 km) hardly disappeared (0.91, a small-to-large ratio of 4.62 against 2.13 outside), while the intermediate craters (20–40 km) were mostly erased (0.47). Moreover, the craters remaining at the centre (40–80 km) are **18% shallower**, but **their rims stand higher** (1.35 against 0.92 in a matched control) — the exact opposite of ordinary erosion or relaxation, which lowers the rim together with the floor!

3. **Two separate foci within the same basin:** the northern focus (the erasure of craters and the landing site of Chang'e-6) sits opposite the heat exits from the near side, while the southern focus (the thinnest crust, the lowest floor and the dense subsurface mass) sits about 450 km to the south of it!

02 · AN IMAGE

An image: a tray of rising dough over a burner flame, and a cake that sinks in the middle after baking

Three simple images explain how a prolonged source of heat beneath the far side — **"the Burner"** — produces all these contradictory phenomena at once:

- **A tray of dough over a burner flame (the rise of the dome and the sliding of the crust to the sides):** imagine soft dough resting over a steady flame: the heat lifts the centre into a dome, and the softened crust slides and is pushed down toward the edges (creating the two thickened crustal rings around SPA). After the flame weakens and the dough cools, the dome collapses inward — and sinks *lower* than its starting height, because the heat drove off volatiles and closed the pores within it!
- **A sheet of wax over a hot plate, and ground packed like wet sand (the shallow craters with high rims):** the heat from below softened and relaxed the ancient craters at the centre (so their floors rose and levelled), and at the same time "baked" and compacted the porous layers of returning material into dense ground. When later craters landed on this baked, compacted

ground, less energy went into compressing pores and more energy built **high rims**!

- **A cake whose edges bake first while its centre sinks (the formation of the outer ring):** when the burner weakened toward the end of the rounding, the edges of the dome cooled and froze first in a raised state (leaving a ridge and a crustal root on the thick eastern side), while the centre stayed hot for a long time, was compacted and finally sank into the deep basin we see today!

03 · SCIENCE AND MODEL

Science and model: the five studies of SPA and the anomalies B1–B6

Part B of Paper 08 presents five full quantitative studies (Studies 1–5, all of which received a full green light) examining the surface, depth, crust, rings and porosity of SPA:

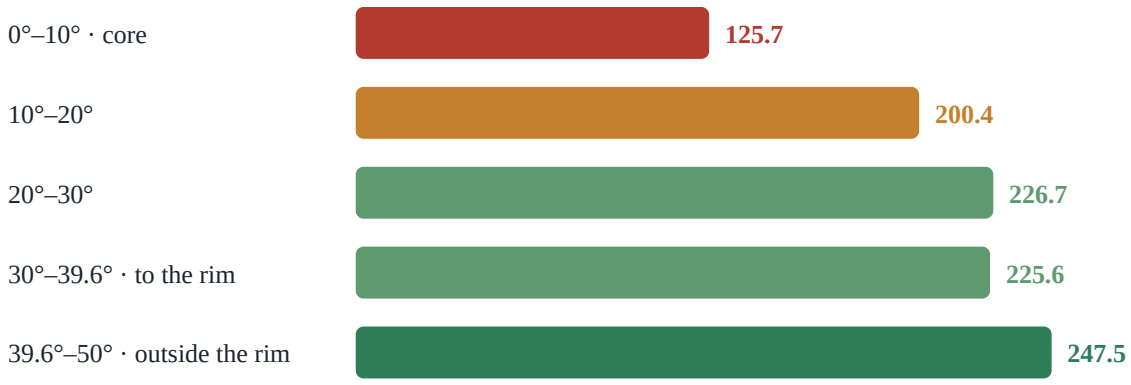
1. **Study 1 — the crater record of SPA (B1):** in a clean count (without lava plains, without the 34 maria and without the 200-km halo), the crater density (20+ km) rises gradually from **125.7 (0.51)** in the 0°–10° ring, to 200.4 (0.81) at 10°–20°, to 226.7 (0.92) at 20°–30°, and up to 247.5 (1.00) outside the rim (39.6°–50°). The anomaly also survives on the ellipse (122–132 in the core against 230–235 outside), and the minimum point (45.4°S, 197.9°E) sits exactly on the meridian of the Serenitatis antipode (198.8°E).

2. **Study 2 — the depth of the basin against the thin crust (B2):** measuring the height on **the plains between the craters** (after removing all craters of 20+ km with their rims to avoid contamination) reveals that the floor of SPA is **1.1–1.8 km** deeper than the isostatic equilibrium of the thin crust alone would give (and with an iron-rich crust at a density of 2,900 kg/m³, the gap grows to **~ 4.4 km**). The depth residual is greatest at the centre (−1.1 to −1.8 km) and fades toward the rim (−0.2 km), and the lowest point (61°S, 199°E, −5.5 km) sits only 35 km from the centre of the dense mass of James et al. (62°S, 200°E).

3 **Study 3 — the outer ring of thickened crust (B4):** after subtracting the regional trend of the far side, a ring of thickened crust appears (3.8–5.7 km at a distance of 50°–55° from the centre, with a crustal root at a depth of 42–56 km) standing above the crust beyond it in seven of eight directions (average +3.9 to +4.4 km), but it is **asymmetric east–west**: thick in the east and north-east (+7.8 and +7.4 km, a one-sided component of 7.7–9.1 km toward 96°–99°) and **completely missing in the west** (−2.3 km in the crust and −0.50 km in height)!

4 **Study 4 — the inner ring and the shift of the centre (B5):** closer to the centre (28°–44°, mean radius 36.7° ≈ 1,114 km) a clear inner ring appears in 22 of 24 directions (especially prominent in the south-east, +24–30 km, and in the west, +6–14 km, where the outer ring is missing!). Its centre sits **108–118 km north-east of the centre of the ellipse** (51.4°S, 196.8°E — the centre of the dome!), and its inner wall is steepest in the north and north-east (30.7 m/km in height and 10.5 km/100 km in the crust, facing the wall of the thick highlands) and gentle in the south-west (11.5 m/km).

5 **Study 5 — the closing of pores at depth (B6):** at the surface, the porosity of the ground in SPA is identical to the highlands (19.7% against 20.3%, reopened by the escape of volatiles). But at a depth of 10 km, **the pores in SPA are almost completely closed at the centre (2.9%–4.4%)** and rise gradually toward the rim (7.1%–8.8%, against 6.2%–12.8% in the rest of the far side). Compacting 12% porosity over a column of 9–15 km explains exactly the sinking of the floor (1.1–1.8 km) found in Study 2!



The density of craters 20 km and more across (per million km²) in rings around the centre of SPA, in a clean count: the core holds only about half the density found outside the rim.

04 · THE MECHANISM IN WORDS

The mechanism in words: an iron-rich floor, little visible lava, buried lava and the thorium crescent (B7–B9)

Parts B7–B9 of Paper 08 link the chemistry and volcanism of the SPA floor to the burner mechanism and to the two-stage history of the central mare:

1. **The iron gradient and the flat potassium-to-thorium ratio (B7):** gamma-ray spectrometer measurements (Lunar Prospector GRS) show that the iron concentration (FeO) falls gradually from **9.48 wt%** at the centre (0°–10°) to 8.56% (10°–20°), 6.63% (20°–30°), and down to 4.43% at the rim — because deep, iron-rich material rose with the heat at the centre of the dome. By contrast, the **K/Th ratio is flat and low (451–455) across the whole width of the basin interior (0°–30°)** (16%–20% lower than the surroundings, 537–567), and jumps only at the rim — because prolonged heating from below drove off the volatile potassium uniformly over the whole heated area!

2. **Why is only 4.1% of SPA's area covered with visible lava despite the thinnest crust? (B8):** in near-side basins with a similarly thin crust (15–17 km, such as Crisium and Humorum), lava covers 22%–74% of the area, while in SPA only **4.1%** (14.6% in the inner 300 km, 8.4% in a central dome the size

of a near-side basin, and 0.7% at the rim). The reason is twofold: first, most of SPA's enormous area is **a zone of heating and melting from below with no eruption at the surface** (which dilutes the area percentage in the denominator); and second, repeated melting emptied the mantle of easily melted elements and made it resistant to later melting (B13).

3. **The buried lava inside the 15° circle and the thorium crescent as a "compass of rotation" at 5.77 hours (B9):** inside the 15° circle around the centre of the dome there lay, at the beginning, **an ordinary large feeding opening (the "central mare")** that fired material outward while the Moon was attached to the Earth and spinning fast (5.77 hours). The regional heat melted and covered this mare — and this is **the buried lava (cryptomare)** that covers **13.6%–24.1%** of the 15° circle (and almost nothing outside it!). And what happened to the thorium-rich material that the central mare fired outward?
    - 4 Measuring 43 craters in the thorium crescent in the north-west (12°–32° from the centre, with peaks of 4.45–4.90 ppm at Birkeland and Oresme V) shows that the thorium concentration is **the same inside the craters and outside them** (32/43 against 34/43) and weakens with distance (r = −0.53) — that is, it is **a cover that landed from above**, not a deep layer dug up from below!
    - 5 **The ballistic trajectory simulation (`sim577.py`):** launching 72 parcels of material at 45° from the tip of the attached Moon at the corotation line (16,400 km, an Earth day of **5.77 hours**) at 0.95–1.0 km/s shows that within a flight of 35–40 minutes the Moon's ground turns forward, and the material lands **on one side only (the trailing side, 74%–96%, asymmetry 0.42–0.54) at a distance of 29°–32°** — exactly the position of the thorium crescent!

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The physical mechanisms of pushing the crust outward by thermal pressure, the advance of the melting front from below (the Stefan condition) and the sinking of the ground through the closing of pores are expressed as follows (§B3, §B6):

$$\Delta p = \alpha K \Delta T \approx 0.4\text{–}1.2 \text{ GPa} \quad (\Delta T = 300 \text{ K})$$

$$\sigma \approx \Delta\rho \, g \, h \approx 5 \text{ MPa}$$

$$v_{\text{melt}} = \frac{q}{\rho \, (c_p \Delta T + L)}$$

$$\Delta z_{\text{compaction}} = \Delta\phi \cdot H$$

For a latent heat L = 4 × 10^5 J/kg, heating by $\Delta T = 800$ K ($c_p \Delta T = 8 \times 10^5$ J/kg) and a crustal density of $\rho = 2,900$ kg/m³, about 7 × 10^{13} J/m² are needed to melt 20 km of crust: even a moderate heat flux of 0.1–0.5 W/m² melts 20 km of crust within only **4.4–22 million years** — far less than the duration of the burner's activity!

06 · THE SIGNATURE

The signature in reality: the grouping of two age systems and the unification of seven observations

The ages measured in the SPA basin divide naturally into two stages in the history of the burner (§B9):

- **The attached stage and the burner's activity (4.25–3.87 Ga):** the crystallisation of the deep norites (4.247 Ga), KREEP-rich basalt (4.203 Ga), the activity of the central mare and the creation of the thorium crescent during the fast 5.77-hour rotation.

- **After the detachment (3.873 Ga) — slow cooling in place with no lateral eruptions:** at the moment of detachment the pressure from the Earth stopped, the central mare calmed, stayed hot and froze slowly in place over about 1.5 billion years (3.7–2.2 Ga, peaking at 3.6–3.2 Ga, and 2.8 Ga at the Chang'e-6 site) — so this young lava has no thorium crescent at all!

The lab box: the profile of craters, depth, crust, porosity and chemistry of SPA (Paper 08, Parts B1–B9)

Ring from the centre of SPA (53°S, 191°E)	Craters ≥ 20 km (clean ratio)	Floor depth against crust alone (km)	GRAIL crustal thickness (model 1 / 4, km)	Porosity at 10 km (exponential / linear)	FeO (wt%) / K/Th ratio
0°–10° (the central core)	**125.7 (0.51)**	**−7.0 against −5.3 to −5.9 (residual −1.1 to −1.8)**	**16.0–16.4 / 22.9–23.4**	**2.9% / 4.4% (surface 19.7%)**	**9.48% / 451**
10°–20°	200.4 (0.81)	−6.4 to −6.5 (residual −0.5 to −1.2)	16.2–17.5 / 22.9–24.2	3.1% / 3.2%	8.56% / 451 (Th peak 1.96)
20°–30°	226.7 (0.92)	−4.0 to −5.1 (residual −0.2 to −0.9)	21.5–26.5 / 28.7–34.2	3.6% / 5.1%	6.63% / 455
30°–39.6° (to the rim)	225.6 (0.91)	−0.8 to −2.7 (residual −0.2 to −0.8)	33.4–39.5 / 42.2–49.5	4.2% / 6.2%	4.91% / 501
39.6°–50° (outside the rim)	**247.5 (1.00)**	**0.0 (reference line)**	**40.6–41.9 / 50.5–51.9**	**7.1% / 8.8%**	**4.43% / 567**

Direction around SPA	Step of the outer ring (45°–60° against 70°–90°)	LOLA height step (km)	Distance of the inner ring and wall steepness (m/km)	Interpretation in the LEF model

North-east (45°)	**+7.4 km (54.8 against 47.4)**	**+1.93 km**	**36° — steepest wall (30.7 m/km)**	**Melting of the thick highland crust and dense piling**
East (90°)	**+7.8 km (46.2 against 38.4)**	**+1.59 km**	**37° — steep wall (17.8 m/km)**	**Maximum one-sided component (7.7–9.1 km toward 97°)**
South (180°)	+6.5 km (37.1 against 30.6)	+1.17 km	42° — gentle slope (14.9 m/km)	Southern crustal root (44 against 29 km)
South-west (225°)	+3.2 km (33.5 against 30.4)	+0.41 km	34° — gentlest slope (**11.5 m/km**)	An ancient thin crust allowing wide spreading
West (270°)	**−2.3 km (36.7 against 39.0)**	**−0.50 km**	**31° — close to the centre (prominent at +4.5 to +13.9 km)**	**No outer ring — the material piled up close by in the inner ring**

07 · NEXT

The door ahead

In this chapter we saw how the subsurface **"burner"** mechanism explains all the phenomena of surface, depth, the two crustal rings, the closing of pores, the little visible lava and the thorium crescent of the SPA basin — and divides the basin into two foci: a northern one (the erasure of craters) and a southern one (the thin crust and the low floor).

Chapter 16

The Depths of SPA, the Chang'e-6 Samples and the Potential Barrier of a Continuous Body →

← Chapter 14 · Prologue

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